

- Introduction

The original website contained five HTML pages and an external CSS file. Part 3 required the addition of an external JavaScript file, search engine optimisation on all pages, new enquiry and contact forms, a sitemap, a robots.txt file, a new Sign Up page and improvements to the CSS code.

- External JavaScript - JS/main.js

The invalid link tag for the external JavaScript file on each of the original pages was replaced with a correct script tag. A new external JavaScript file (JS/main.js) was created that contained all of the JavaScript for the website. No JavaScript code existed inside any of the HTML files. The following features were added to the website through the use of JavaScript:

Fade-in effect to all pages on load

A scroll progress bar in the header

A glow on the header when scrolling down the page

A mobile hamburger menu

Automatic selection of the active page in the navigation

Scroll reveal animations for fade in elements

A counter for stats on the About page that counts up when scrolled into view

A live search bar for products

A lightbox for product images

A news ticker on the home page

Enquiry form with dynamic responses

Contact form that opens the user's email application with the message pre-populated

A four stage sign up form

A back to top button

The current year is automatically added to the footer

Smooth scrolling for all links on the website

- Search Engine Optimisation

The original website had no elements for search engine optimisation. On each web page the following elements were added:

Meta description

Keywords

Author

Index and follow robots directive

Open Graph meta tags

All images have new and improved alt text

A robots.txt file has been created

A sitemap has been created

- Forms Added

No forms existed on the original website. The following forms were added:

An enquiry form and response on the Enquiry page

A contact form on the Contact Details page that opens the user's email application with the contact information already typed in

- New Pages Created

An Enquiry page was created with a form on the left and information on the right. A Sign Up page was created with a banner at the top of the page and three main features with a four stage sign up form.

- CSS Improvements

A total of fifteen colour variables were defined instead of the original four. New variables for type sizes and spacing were created. All of the new features created through JavaScript had their styles added to the CSS. The background colour of all input fields was previously the default white of the user's browser. This was corrected with the use of the color-scheme dark property, a forced dark background to the site and a webkit autofill shadow to the input fields. These styles were added to the stylesheet and an inline style block in each of the HTML files.

- Changes to Existing Pages

The link to the external JavaScript file on each page was replaced. The navigation on each page was changed to include links to the Enquiry and Sign Up pages. Meta tags for SEO were added to each page. The inline style block was added to each page to change the colour of the input fields.

The Home page also received a news ticker, a series of info boxes and a few call to action buttons.

The Product page received a search bar for the products. The incorrect image filename Prodcuts was changed to Products.

The Sessions page had the scope attribute added to the table headers.

The Contact Details page had social media links added including the official Facebook logo, a link that will work on mobile phones to call the contact number, an email link that can be tapped on mobile phones and a contact form.

- Bug Fixes from Part 2

The copyright symbol that used a colon was changed to a semicolon on all pages.

A stray quotation mark was removed from the About page.

The image on the About page had its width of 1500px removed to avoid a horizontal scroll bar on smaller screens.

Apostrophes were added to the names of the organisation and the text “don’t” on the About page.

The product image filename that said Prodcuts was changed to Products.

The footer on the Product page was moved into the body tag.

The unclosed font tag was removed from the Sessions page.

A stray quotation mark was removed from the Contact Details page html tag.

The Google maps element was moved out of the unordered list on the Contact Details page.

All inline attributes and tags for colour, align, background and font were removed from the website

The Aptos font was replaced with Google Fonts: Bebas Neue, Oswald and Lato

References

Facebook for Developers (2024) Brand Resources and Logo Usage. Available at: <https://developers.facebook.com/docs/sharing/webmasters/> (Accessed: 18 June 2026).

Google (2024) Google Maps Embed API. Available at: <https://developers.google.com/maps/documentation/embed/get-started> (Accessed: 18 June 2026).

Google Fonts (2024) Bebas Neue, Oswald, Lato font families. Available at: <https://fonts.google.com> (Accessed: 18 June 2026).

Google Search Central (2024) Introduction to robots.txt. Available at: <https://developers.google.com/search/docs/crawling-indexing/robots/intro> (Accessed: 18 June 2026).

Google Search Central (2024) Open Graph meta tags. Available at:
<https://developers.google.com/search/docs/appearance/structured-data/article> (Accessed: 18 June 2026).

MDN Web Docs (2024) Client-side form validation. Available at:
https://developer.mozilla.org/en-US/docs/Learn/Forms/Form_validation (Accessed: 18 June 2026).

MDN Web Docs (2024) Intersection Observer API. Available at:
https://developer.mozilla.org/en-US/docs/Web/API/Intersection_Observer_API (Accessed: 18 June 2026).

MDN Web Docs (2024) The anchor element and mailto links. Available at:
<https://developer.mozilla.org/en-US/docs/Web/HTML/Element/a> (Accessed: 18 June 2026).

Sitemaps.org (2024) XML Sitemap protocol specification. Available at:
<https://www.sitemaps.org/protocol.html> (Accessed: 18 June 2026).

W3C (2021) Web Content Accessibility Guidelines (WCAG) 2.1. Available at:
<https://www.w3.org/TR/WCAG21/> (Accessed: 18 June 2026).

W3Schools (2024) HTML Forms tutorial. Available at:
https://www.w3schools.com/html/html_forms.asp (Accessed: 18 June 2026).

W3Schools (2024) JavaScript HTML DOM EventListener. Available at:
https://www.w3schools.com/js/js_html_dom_eventlistener.asp (Accessed: 18 June 2026).